

Project Initialization and Planning Phase

Date	28 June 2024
Team ID	
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of the Smart Lender – Loan Eligibility Prediction System is to develop an intelligent machine learning-based web application that predicts loan eligibility using applicant information. The system aims to automate the loan evaluation process, reduce manual effort, improve prediction accuracy, and provide faster decision-making for banks and financial institutions.
Scope	The project focuses on collecting and preprocessing loan applicant data, training multiple machine learning classification models, selecting the best-performing model (XGBoost), and integrating it into a Flask-based web application . The system enables users to submit loan details and instantly receive loan approval or rejection predictions through a user-friendly interface.
Problem Statement	

Impact	An intelligent loan eligibility prediction system improves operational efficiency, reduces processing time, minimizes credit risk, enhances decision-making accuracy, and provides customers with faster and more transparent loan approval services.
Proposed Solution	
Approach	Develop a machine learning-based loan eligibility prediction system that preprocesses applicant data, trains multiple classification algorithms (Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost), selects the best-performing model, and deploys it through a Flask web application to provide real-time loan eligibility predictions.
Key Features	1. Automated data preprocessing and feature engineering. 2. Comparison of multiple classification algorithms. 3. XGBoost model for high prediction accuracy. 4. Flask backend for handling user requests and predictions. 5. Interactive HTML, CSS, and JavaScript frontend. 6. Machine learning-based loan eligibility prediction. 7. Real-time loan approval or rejection prediction. 8. Cloud deployment using IBM Cloud.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 or above
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	512 GB SSD or above
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv